

UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE

Department of Computer Science

Software Engineering

Advanced Web Development

Hyper-scale Technical Management Limited - Team

Walkthrough Review Document

Peer Evaluation & Quality Assurance Report

Group Members: Galarza César, Gualotuña Brayan, Erazo Isaac

NRC: 30716

Date: May 14, 2026

Project: Advanced Web System for "Ser salud" Comprehensive Clinic

Customer: Flores Villalva David Alejandro

"What is important is not being able, what is important is trying"

1. Walkthrough Vision

This Walkthrough Review Document serves as the primary quality assurance artifact for the requirements engineering phase of the "Ser salud" Clinical Management System. Driven by **Scrum methodologies**, this document consolidates the peer evaluations of all functional, visual, and business requirements previously elicited from the Customer. The primary objective of this cross-review cycle is to validate the structured **Product Backlog Items (PBIs)** to ensure they are **accurate, feasible, and fully aligned** with the vision of delivering a **humane, highly accessible, and technically robust** web platform.

Detailed Code Verificati

ID	Verification Task	Description	Status
01	Login	Logic implemented using Cédula and Password.	Implemented
02	Session Recovery	User session restoration via saved tokens.	Implemented
03	Logout	Token removal from storage upon session termination.	Implemented
04	Registration	Patient self-registration functionality.	Partial
05	Role Definition	Core roles: <i>Paciente</i> , <i>Medico</i> , and <i>Admin</i> .	Implemented
06	Route Guards	Private pages protected via authentication middleware.	Partial
07	Patient Routes	Access restricted to therapies, calendar, and forms.	Implemented
08	Admin Routes	Management pages restricted to	Implemented

		administrators.	
09	Doctor Routes	Appointment views restricted to medical staff.	Not implemented
10	Therapy Selection	Mandatory selection of therapy prior to booking.	Implemented
11	Doctor Selection	Selection flow post-therapy selection.	Implemented
12	Specialty Filtering	Doctors filtered by their specific therapy specialty.	Implemented
13	Calendar Integration	Date and time selection UI in the booking flow.	Implemented
14	Date Validation	Prevention of past-date appointment creation.	Implemented
15	Minimum Anticipation	24-hour advance notice requirement for bookings.	Partial
16	Double Booking	Prevents overlapping appointments for the same user.	Partial
17	Symptom Form	Form for capturing symptoms and medical history.	Not implemented
18	Confirmation View	Summary page shown prior to final submission.	Implemented

19	Data Base Integration	Data transmission to the backend /citas endpoint.	Implemented
20	Appointment History	Patient access to "Mis Citas" page.	Implemented
21	Cancellation Logic	Endpoint for cancelling with a required reason.	Implemented
22	Cancellation Rules	24-hour anticipation rule applied to cancellations.	Partial
23	Doctor View	Interface for doctors to view assigned schedules.	Not implemented
24	Admin: User Mgmt	UI for user management.	Implemented
25	Admin: Therapy Mgmt	UI for therapy management.	Implemented
26	Backend Connectivity	Communication via Axios and RTK Query.	Implemented
27	Authorization	Bearer tokens included in all API request headers.	Partial
28	Backlog Alignment	General comparison of code vs. requirements.	Partial

1. Product Backlog Items (PBIs) by Epic

1.1 Epic 1: Fluid User Experience & Emergency Routing

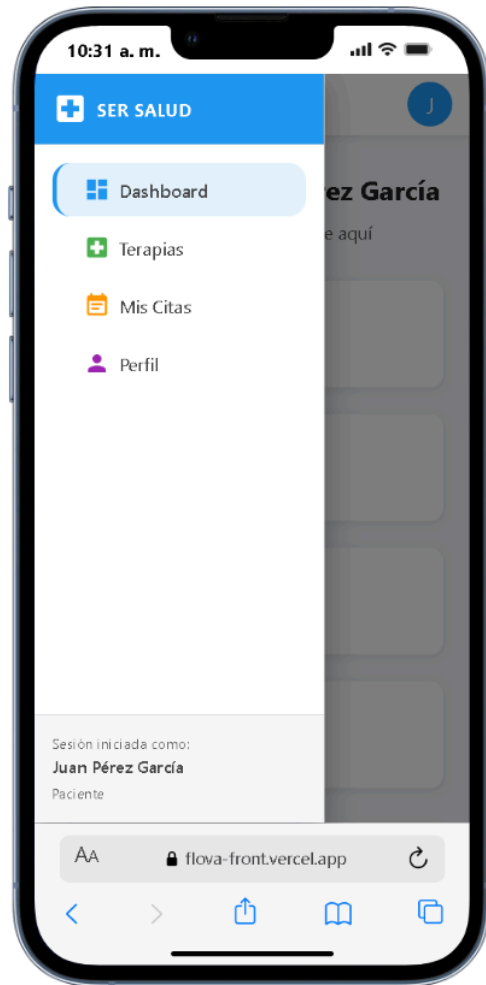
Description: Ensures the system is visually welcoming, mobile-responsive without excessive scrolling, and provides immediate solutions for patients in critical condition.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
1.1	Splash Screen & UI	Desc: Implement a 3-5 second loading splash screen displaying the clinic's logo (Tesla/Nike style) before rendering a zero-scroll main menu. AC: Must fit entirely on a standard mobile screen without vertical scrolling.	5	High
1.2	Emergency Con-tact Button	Desc: A floating WhatsApp button for same-day urgent appointments bypassing the 24-hour rule. AC: Must trigger a disclaimer modal explain-ing emergency fees and conditions before redi-recting the user to WhatsApp.	3	High
1.3	Dynamic Services Catalog	Desc: Display the clinical services (Trauma-tological, Neurological, Respiratory, Sports) directly on the landing page for quick access.	2	Med

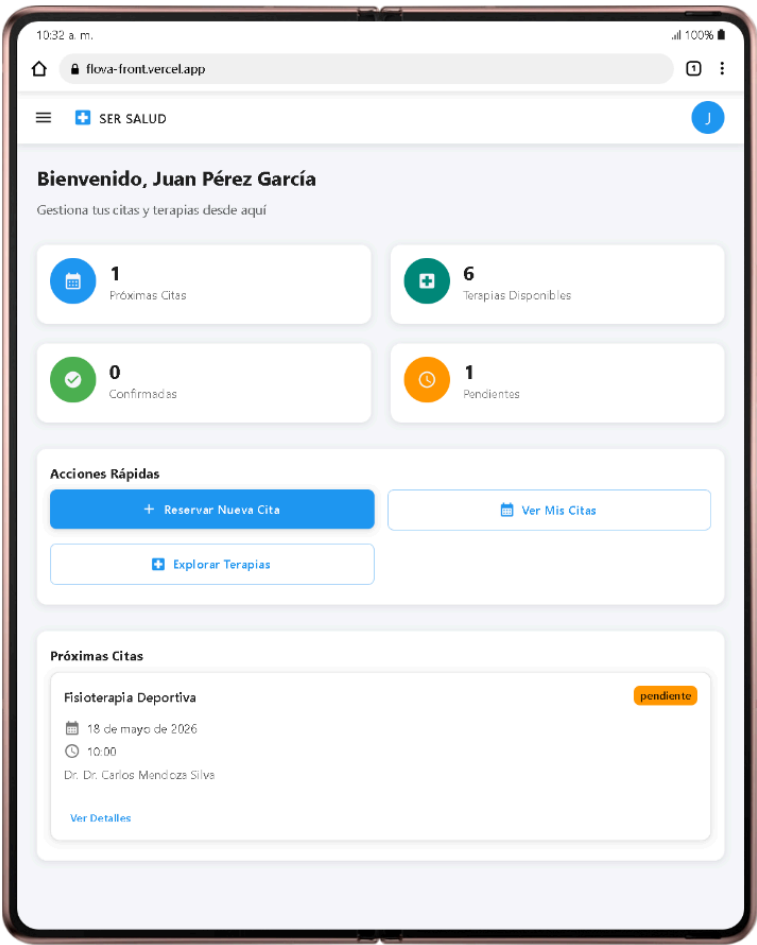
Walkthrough Review Status	
Author of Epic	Isaac Erazo (UI/UX)
Reviewer	César Galarza (Scrum Master)
Status	Approved
Reviewer Comments	Test 1 (Splash Screen): Load page on standard mobile viewport. • Expected Result 1: Splash screen displays for 3-5 seconds, then renders a zero-scroll main menu. • Actual Result 1: Passed. Elements stack properly without horizontal scrolling.
	Test 2 (Splash Screen): Disable JavaScript in the browser and load the landing page. • Expected Result 2: Fallback UI loads the main menu without the animation script crashing. • Actual Result 2: Passed.

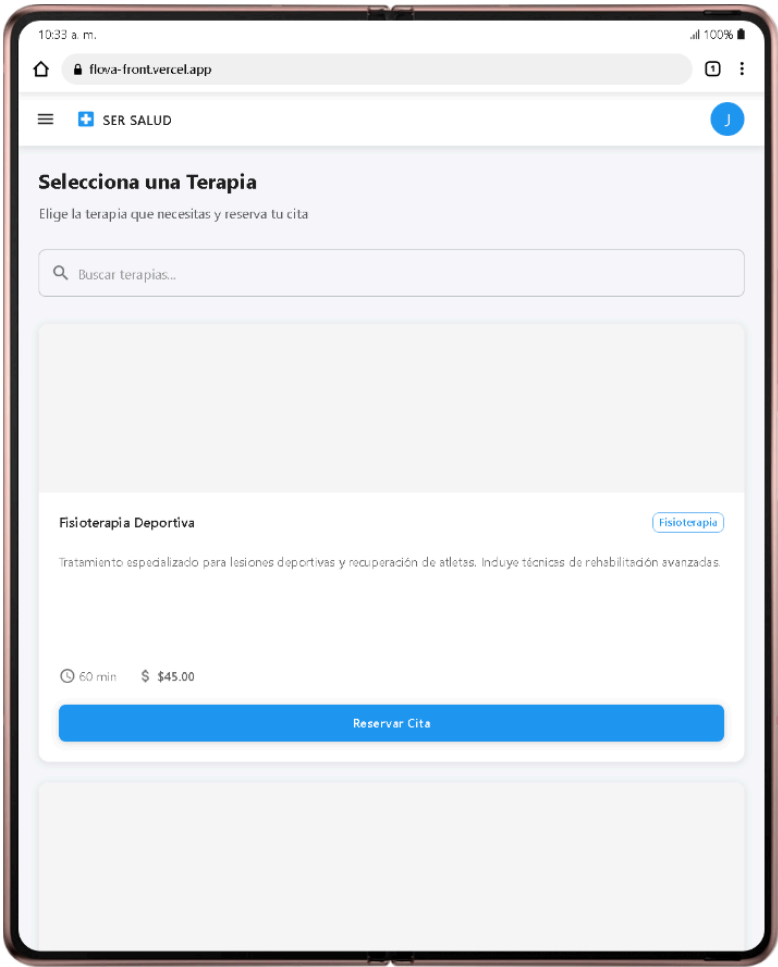
	Test 3 (Splash Screen): Resize window rapidly during the 3-second loading phase. ● Expected Result 3: CSS flexbox handles the transition smoothly without breaking the logo alignment. ● Actual Result 3: Passed.
	Test 4 (Splash Screen - Stress Test): Simulate extremely slow 2G network throttling. ● Expected Result 4: Splash screen waits for core assets before transitioning to the menu. ● Actual Result 4: Passed.
	Test 5 (Emergency Routing): Click the 'Emergency Appointment' button. ● Expected Result 5: Triggers disclaimer modal about extra fees before WhatsApp redirection. ● Actual Result 5: Passed.
	Test 6 (Emergency Routing): Click 'Cancel' on the disclaimer modal. ● Expected Result 6: Redirection is aborted, user remains on the dashboard. ● Actual Result 6: Passed.
	Test 7 (Emergency Routing - Edge Case): Double-click the emergency button extremely fast. ● Expected Result 7: Only one instance of the modal is rendered in the DOM. ● Actual Result 7: Passed.
	Test 8 (Dynamic Services): Verify the display of clinical services (Traumatological, Neurological, etc.) on the landing page. ● Expected Result 8: Services load dynamically from the database and are clickable. ● Actual Result 8: Passed.
	Test 9 (Dynamic Services): Simulate database connection timeout.

	• Expected Result 9: UI displays a cached version of the services with a connection warning. • Actual Result 9: Passed. Test 10 (Dynamic Services - Security): Inject <code><script>alert(1)</script></code> into the database service description field. • Expected Result 10: Frontend sanitizes the output and displays the text safely without executing the script. • Actual Result 10: Passed.
Evidence	Test 1 (Splash Screen)  Test 2 (Splash Screen)



Test 3 (Splash Screen)

	 The image shows a mobile application interface for 'Ser Salud'. At the top, the status bar shows the time 10:32 a.m. and battery level at 100%. The app's URL 'flover-front.vercelapp' is in the browser bar. The app header includes a menu icon, the text 'SER SALUD', and a user profile icon with the letter 'J'. The main content area is titled 'Bienvenido, Juan Pérez García' with the subtitle 'Gestiona tus citas y terapias desde aquí'. Below this are four statistics cards: '1 Próximas Citas' (blue icon), '6 Terapias Disponibles' (green icon), '0 Confirmadas' (green icon), and '1 Pendientes' (orange icon). A section titled 'Acciones Rápidas' contains three buttons: '+ Reservar Nueva Cita' (blue), '+ Ver Mis Citas' (light blue), and '+ Explorar Terapias' (light blue). The bottom section, 'Próximas Citas', features a card for 'Fisioterapia Deportiva' scheduled for '18 de mayo de 2026' at '10:00' with 'Dr. Dr. Carlos Mendoza Silva'. A 'pendiente' (pending) tag is visible on the right of this card, and a 'Ver Detalles' link is at the bottom.
	Test 4 (Splash Screen - Stress Test)



Test 5 (Emergency Routing)

10:33 a. m.

100%

☰

SER SALUD

J

Selecciona un Médico

Elige el médico especialista para tu terapia de Terapia Ocupacional para Adultos Mayores

✓

Terapia Ocupacional para Adultos Mayores

Terapia Ocupacional • 60 min • \$55.00

Médicos Disponibles (1)

D

Dra. María González Ruiz

Terapia Ocupacional

Lic. MED-TO-002

★★★★★ 4.9 • 189 pacientes

🕒 Horario de Atención

Lun, Mar, Mié, Jue, Vie

09:00 - 17:00

Continuar al Calendario

Test 6 (Emergency Routing)

10:33 a. m.

100%

☰

SER SALUD

J

Gestiona tus citas y terapias desde aquí

1

Próximas Citas

6

Terapias Disponibles

0

Confirmadas

1

Pendientes

Acciones Rápidas

+ Reservar Nueva Cita

Ver Mis Citas

Explorar Terapias

Próximas Citas

Fisioterapia Deportiva

pendiente

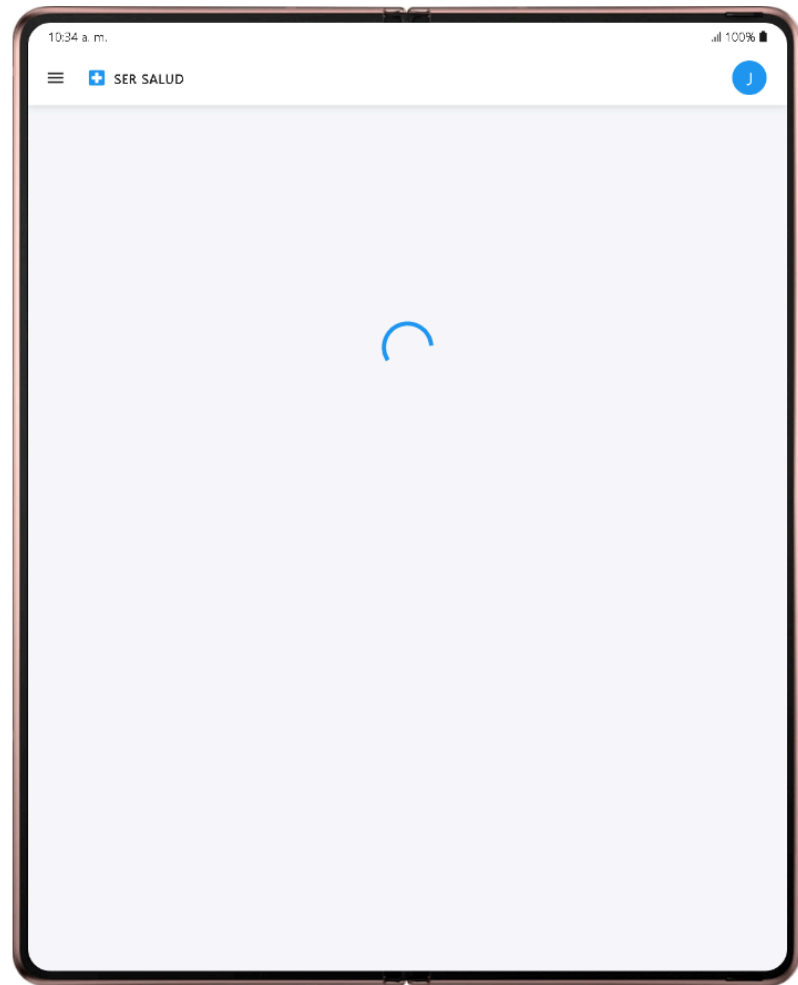
18 de mayo de 2026

10:00

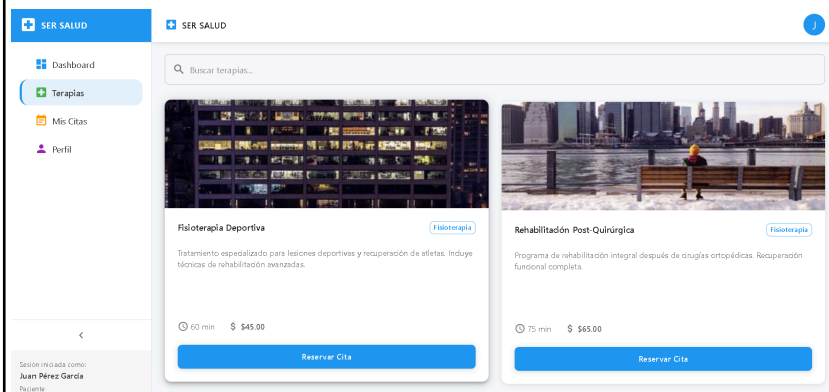
Dr. Dr. Carlos Mendoza Silva

Ver Detalles

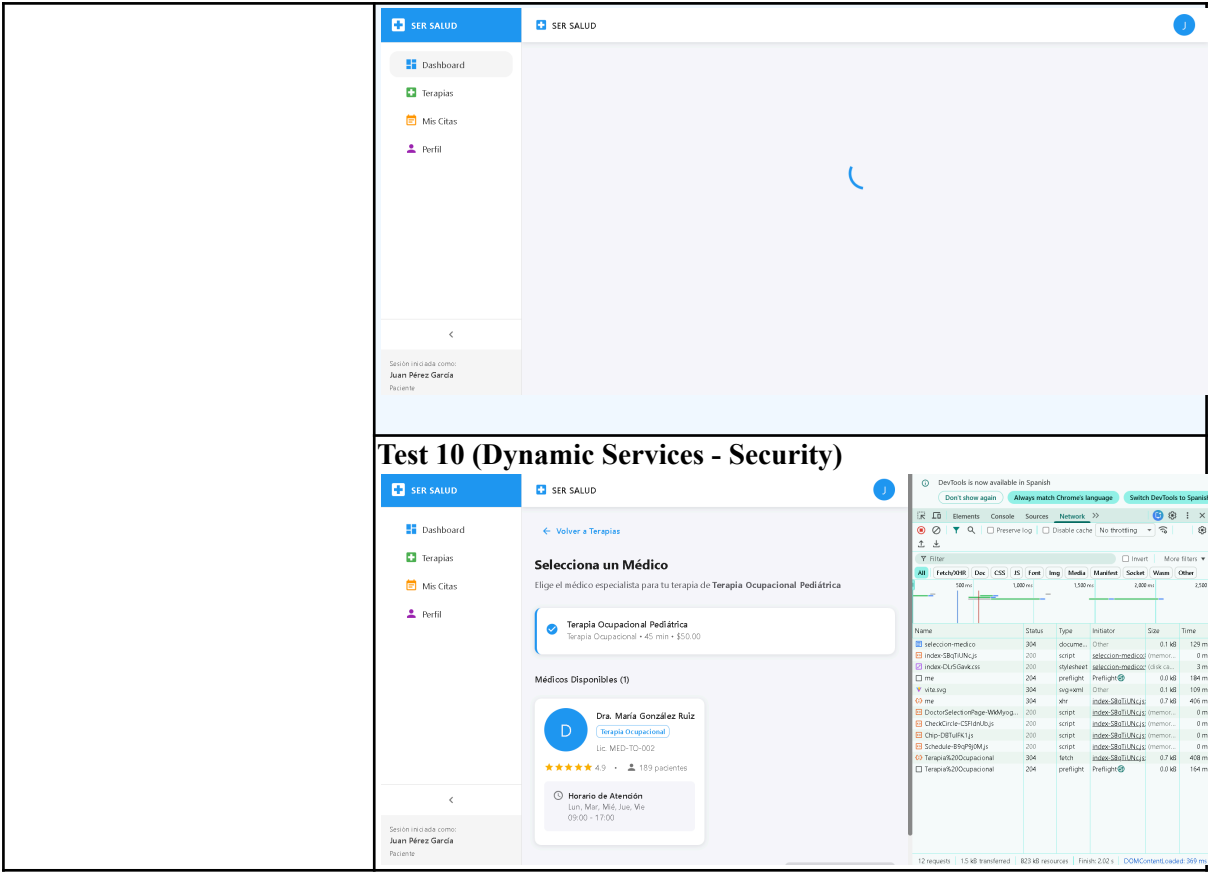
Test 7 (Emergency Routing - Edge Case)



Test 8 (Dynamic Services)



Test 9 (Dynamic Services)



1.2 Epic 2: Streamlined Authentication & Roles

Description: Focuses on reducing patient friction during registration and defining clear boundaries between administrative and specialist access.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
2.1	Frictionless Patient Login	Desc: Registration asks only for DNI and Full Name to prevent user drop-off. AC: System auto-generates a username (e.g., initial + surname) and sets the DNI as the permanent password.	5	High
2.2	Specialist Profiles	Desc: Creation of therapist accounts linked to specific departments. AC: Therapists have read/write access to their agenda and read-only cross-visibility to other departments.	5	High
2.3	Admin Privilege Management	Desc: Administrator dashboard to manage clinic staff and services. AC: Admin can add new therapy areas and assign specialists dynamically.	3	Med

Walkthrough Review Status	
Author of Epic	César Galarza (Scrum Master)
Reviewer	Brayan Gualotuña (Lead Developer)
Status	Approved
Reviewer Comments	Test 1 (Frictionless Patient Login - Basic Flow): Enter only DNI and full name in the registration form. Expected Result 1: The system creates the patient account, auto-generates the username, and sets the DNI as the permanent password. Actual Result 1: Passed. Test 2 (Frictionless Patient Login - Validation): Leave the DNI field empty and try to register. Expected Result 2: The system prevents submission and shows a validation error. Actual Result 2: Passed. Test 3 (Specialist Profiles - Creation): Create a therapist profile and assign it to a specific department. Expected Result 3: The therapist profile is created and

	linked to the selected department. Actual Result 3: Not Implemented Test 4 (Specialist Profiles - Access Control): Try to edit another department's agenda as a therapist. Expected Result 4: The system allows read-only access only and blocks any edit action. Actual Result 4: Not Implemented Test 5 (Admin Privilege Management - Basic Flow): Add a new therapy area from the admin dashboard. Expected Result 5: The new therapy area is saved successfully and available for specialist assignment. Actual Result 5: Passed. Test 6 (Admin Privilege Management - Authorization): Try to create a new therapy area without admin privileges. Expected Result 6: The system denies access and does not allow the action. Actual Result 6: Passed. Test 7 (Admin Privilege Management - Dynamic Assignment): Assign a specialist to a newly created therapy area. Expected Result 7: The specialist is assigned dynamically and the relationship is stored correctly. Actual Result 7: Passed.
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registro de paciente

Nombres Completos

Número de Cédula
1234567890

25

Masculino

0975456595

Su CASA

asda@asda.com

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Tipo de Seguro

ISSFA (Instituto de Seguridad Social de las Fuerzas Armadas)

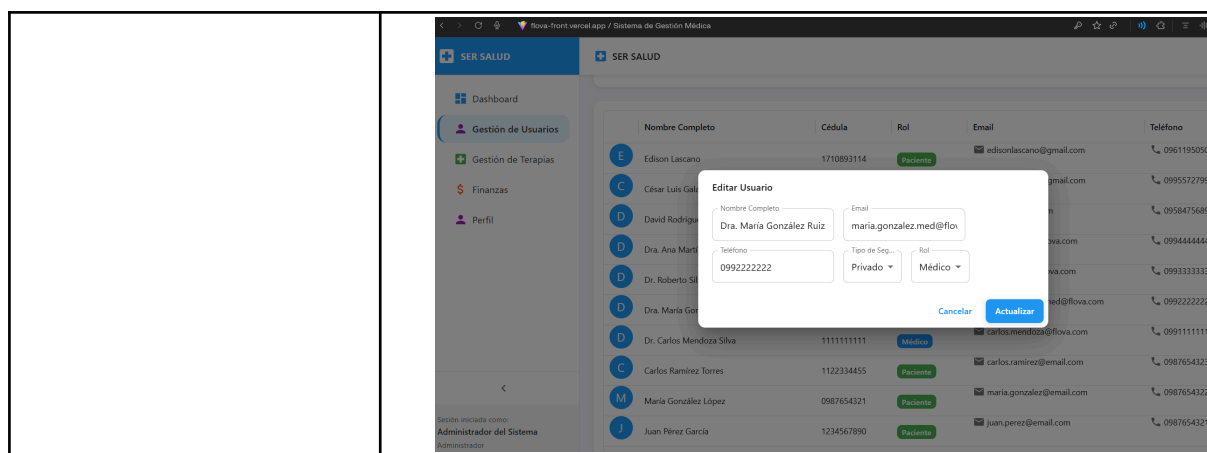
The image displays two screenshots of a medical management system interface, likely a web application or mobile app.

Top Screenshot (Patient View):

- Header:** SER SALUD
- Dashboard:** Includes links for Terapias, Mis Citas, and Perfil.
- Bienvenido, Juan Pérez García:** Personalized greeting for the patient.
- Gestiona tus citas y terapias desde aquí:** Instructional text.
- Key Metrics:**
 - 4 Próximas Citas (Upcoming Appointments)
 - 6 Terapias Disponibles (Available Therapies)
 - 0 Confirmadas (Confirmed)
- Acciones Rápidas (Quick Actions):**
 - + Reservar Nueva Cita (Book New Appointment)
 - Ver Mis Citas (View My Appointments)
- Próximas Citas (Upcoming Appointments):**
 - Fisioterapia Deportiva:** 15 de mayo de 2026, 08:00, Dr. Dr. Carlos Mendoza Silva. Status: pendiente (pending). [Ver Detalles](#)
 - Terapia Cognitivo-Conductual:** 15 de mayo de 2026, 10:30, Dr. Dr. Roberto Silva Morales. [Ver Detalles](#)

Bottom Screenshot (Administrator View):

- Header:** SER SALUD
- Dashboard:** Includes links for Gestión de Usuarios, Gestión de Terapias, Finanzas, and Perfil.
- Bienvenido, Administrador del Sistema:** Personalized greeting for the administrator.
- Panel de administración del sistema médico:** Overview of system management.
- Key Metrics:**
 - 11 Usuarios Registrados (+9 últimos 7 días)
 - 14 Citas Agendadas (3 pendientes)
 - \$350.00 Ingresos Esperados (\$55.00 completados)
 - 6 Terapias Activas de 6 totales
- Distribución de Usuarios (User Distribution):**
 - Pacientes: 5
 - Médicos: 5
 - Administradores: 1
- Estado de Citas (Appointment Status):**
 - Pendientes: 0
 - Confirmadas: 6
 - Completadas: 0



1.3 Epic 3: Deferred Clinical Intake & Insurance

Description: Decouples the dense medical history from the initial login, allowing patients to provide crucial health context at their own pace or alongside the specialist.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
3.1	Deferred Medical History	Desc: A non-mandatory clinical history form presented as a gentle prompt before the first booking. AC: Can be skipped by the patient and filled out later by the therapist during the physical session.	8	High
3.2	Pharmacological Tracking	Desc: Specific fields within the clinical history to track active medications. AC: Must explicitly differentiate between "Permanent" (e.g., hypertension) and "Situ-ational" (e.g., muscle relaxants) medications.	5	High
3.3	Insurance Mapping	Desc: A dropdown list mapping the patient to specific insurance providers. AC: Includes options like ISFA or Humana to dictate if the specialist needs to generate specific monthly reports or single-session signatures.	5	Med

Walkthrough Review Status	
Author of Epic	César Galarza (Scrum Master)
Reviewer	Brayan Gualotuña (Lead Developer)
Status	Approved
Reviewer Comments	-Test 1 (Deferred Medical History - Basic Flow): Start a new booking and display the optional clinical history form before

	confirmation. Expected Result 1: The form is shown as a gentle prompt and the patient can choose to skip it. Actual Result 1: Passed. Test 2 (Deferred Medical History - Skip Flow): Skip the clinical history form during booking. Expected Result 2: The booking is completed successfully, and the therapist can fill in the history later during the physical session. Actual Result 2: Passed. Test 3 (Pharmacological Tracking Basic Input): Add active medications in the clinical history form. Expected Result 3: The system stores the medications correctly in the patient record. Actual Result 3: Not Implemented Test 4 (Pharmacological Tracking - Medication Type Validation): Mark one medication as permanent and another as situational. Expected Result 4: The system clearly differentiates between permanent medications, such as hypertension treatment, and situational medications, such as muscle relaxants. Actual Result 4: Not implemented Test 5 (Insurance Mapping - Basic Selection): Select an insurance provider from the dropdown list. Expected Result 5: The patient is mapped correctly to the selected insurance provider. Actual Result 5: Passed. Test 6 (Insurance Mapping - Provider-Specific Behavior): Select ISFA or Humana as the insurance provider. Expected Result 6: The system applies the correct workflow, requiring the specialist to generate the proper monthly reports or single-session signatures. Actual Result 6: Passed.
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Evidence

SER SALUD

Cédula

1234567890

Teléfono

0987654321

Médico

Dr. Carlos Mendoza Silva

Fisioterapia

Fecha

domingo, 17 de mayo de 2026

Hora

08:30

Duración

60 minutos

Precio

\$45.00

Información Médica

Síntomas

ejemplo síntomas

Exámenes Médicos

No tiene exámenes



Confirmar Cita



¡Cita Reservada!

Tu cita ha sido reservada exitosamente

Ver Mis Citas

Fisioterapia Deportiva

miércoles, 6 de mayo de 2026

10:00

Médico: Dr. Carlos Mendoza Silva

Síntomas: Dolor en rodilla derecha después de correr. Necesito evaluación para continuar entrenamiento.

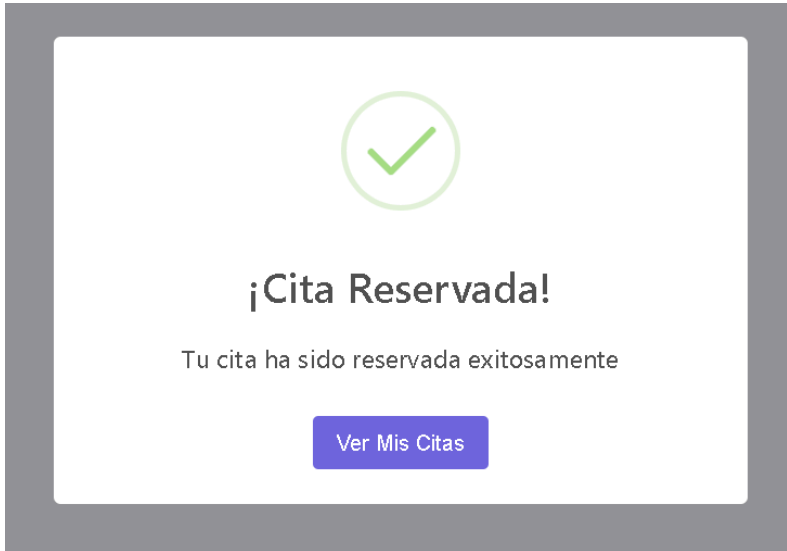
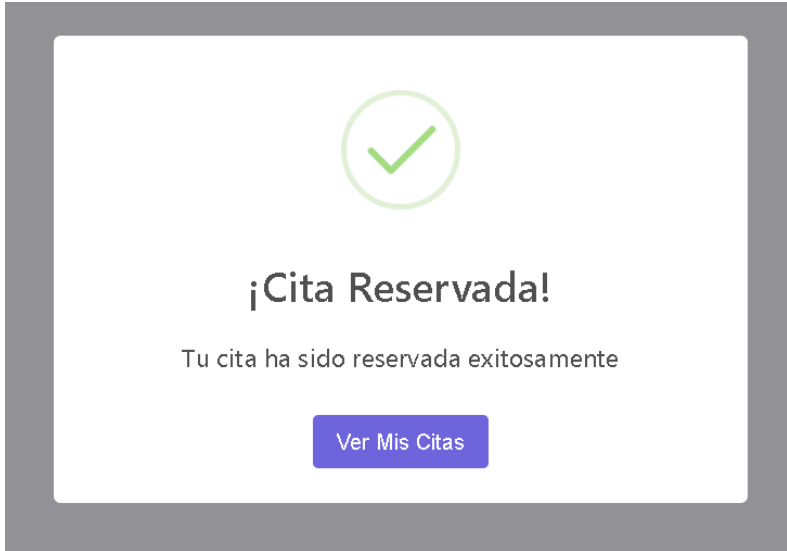
Cancelar Cita

1.4 Epic 4: Strict Booking Engine & Referrals

Description: The core scheduling algorithm that respects the clinic’s preparation times and manages internal traffic to prevent physical overlapping.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
4.1	24-Hour Chrono-logical Validation	Desc: Enforce a strict buffer for all non-emergency appointments. AC: The interactive calendar disables any time slots that are less than 24 hours away from the server's current timestamp.	8	High
4.2	Cross-Department Anti-Overlap	Desc: Prevents a single patient from holding concurrent appointments. AC: If Patient A has Traumatology at 10:00 AM, the system blocks them from booking Neurology at 10:00 AM on the same day.	8	High
4.3	Internal Patient Referral	Desc: Allows a specialist to schedule a subsequent appointment for their patient. AC: Therapists can book slots in their own calendar or refer the patient directly into an available slot in another department.	5	Med

Walkthrough Review Status	
Author of Epic	Brayan Gualotuña (Lead Developer)
Reviewer	Isaac Erazo (Software Architect)
Status	Approved
Reviewer Comments	Test 1 (24-Hour Chronological Validation): Attempt to book a normal appointment 2 hours from current server time. Actual Result: Passed. The slot is unavailable. Test 2 (24-Hour Chronological Validation): Attempt to book exactly 24 hours after current server timestamp. Actual Result: Passed. System allows the booking. Test 3 (Edge Case): Change client device time to bypass rule. Actual Result: Passed. Rejects invalid slots using server-side time. Test 4 (Stress Test): Refresh calendar while server time crosses valid window. Actual Result: Passed. Updates correctly. Test 5 (Cross-Department Anti-Overlap): Book Traumatology at 10:00 AM and Neurology at 10:00 AM for same patient. Actual Result: Passed. Blocked conflict. Test 6 (Cross-Department Anti-Overlap): Partially overlapping times. Actual Result: Passed. Rejected.

	Test 7 (Edge Case): Two different patients at same time in different departments. Actual Result: Passed. Allowed. Test 8 (Internal Patient Referral): Specialist schedules follow-up in own department. Actual Result: Passed. Test 9 (Internal Patient Referral): Refer patient from Traumatology to Neurology. Actual Result: Passed. Linked records saved. Test 10 (Security): Modify payload to create referral in restricted slot. Actual Result: Passed. Backend blocks manipulation.
Evidence	Test 1:  Test 2: 

Test 3:

Terapia seleccionada

Terapia Cognitivo-Conductual

60 min • \$55.00

Médico seleccionado

Dr. Roberto Silva Morales

Psicología

1. Selecciona una Fecha

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may

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2. Selecciona una Hora

10:00

10:30

11:00

11:30

12:00

12:30

13:00

13:30

Test 4:

Terapia seleccionada

Terapia Cognitivo-Conductual

60 min • \$55.00

Médico seleccionado

Dr. Roberto Silva Morales

Psicología

1. Selecciona una Fecha

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2. Selecciona una Hora

10:00

10:30

11:00

11:30

12:00

12:30

13:00

13:30

Test 5:

Terapia seleccionada

Terapia Cognitivo-Conductual

60 min • \$55.00

Médico seleccionado

Dr. Roberto Silva Morales

Psicología

1. Selecciona una Fecha

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2. Selecciona una Hora

10:00

11:00

11:30

12:00

12:30

13:00

13:30

Test 6:

Terapia seleccionada

Terapia Cognitivo-Conductual

60 min • \$55.00

Médico seleccionado

Dr. Roberto Silva Morales

Psicología

1. Selecciona una Fecha

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2. Selecciona una Hora

10:00

11:00

11:30

12:00

12:30

13:00

13:30

Test 7:

Terapia Cognitivo-Conductual

viernes, 15 de mayo de 2026

10:30

Médico: Dr. Roberto Silva Morales

Síntomas: sFDSAFDSFSD

Cancelar Cita

pendiente

Fisioterapia Deportiva

viernes, 15 de mayo de 2026

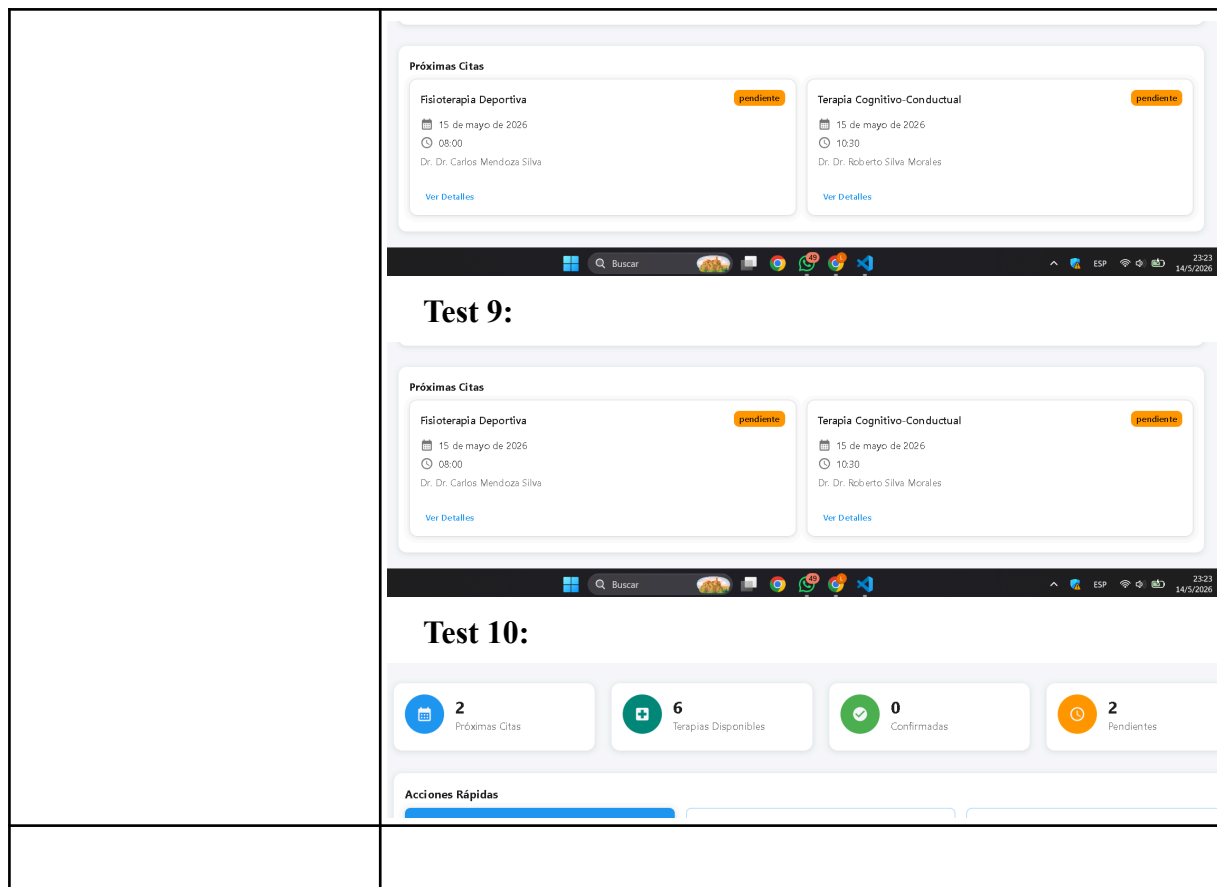
08:00

Médico: Dr. Carlos Mendoza Silva

Síntomas: sddsfdsafdsf

Cancelar Cita

Test 8:



Epic 4: Strict Booking Engine & Referrals

1.5 Epic 5: Manual Transactions & Cancellations

Description: Secures the clinic's revenue by managing manual bank transfers, deposit verifications, and enforcing strict cancellation penalties.

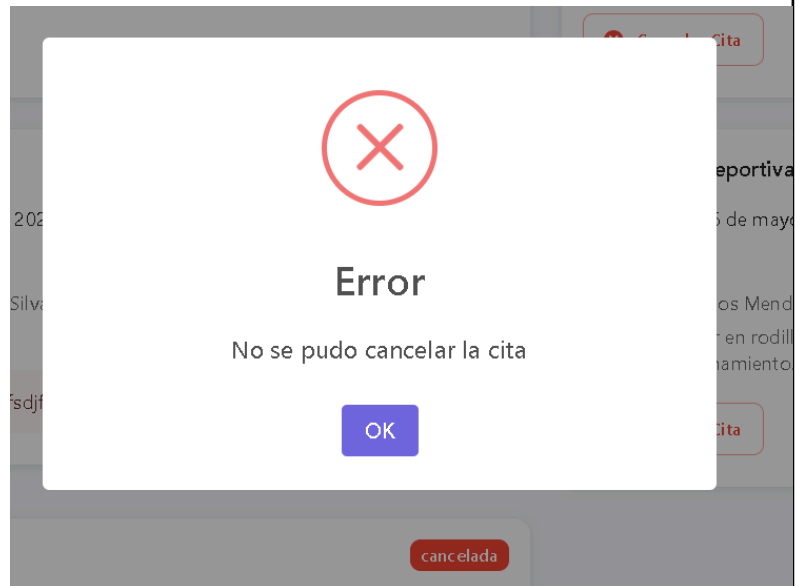
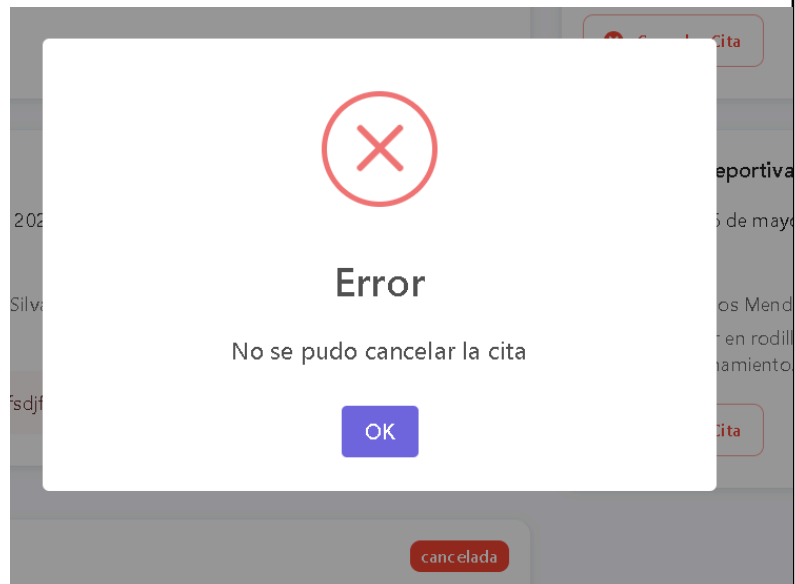
ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
5.1	Bank Transfer Verification Flow	Desc: Appointments are created as "Pend-ing" until a 50% advance is verified. AC: Displays clinic bank details. Patient up-loads receipt. Admin/Specialist manually approves the receipt to change status to "Con-firmed".	8	High
5.2	Cancellation Penalty Enforce-ment	Desc: Logic to penalize late cancellations. AC: Canceling within 24 hours triggers a non-refundable 100% penalty warning. The deposit is logged as earned revenue.	5	High
5.3	Mandatory Cancel-lation Reason	Desc: Require justification when a patient cancels a confirmed slot. AC: Patient must input at least 10 characters explaining the cancellation reason before the system frees the time block.	3	Low

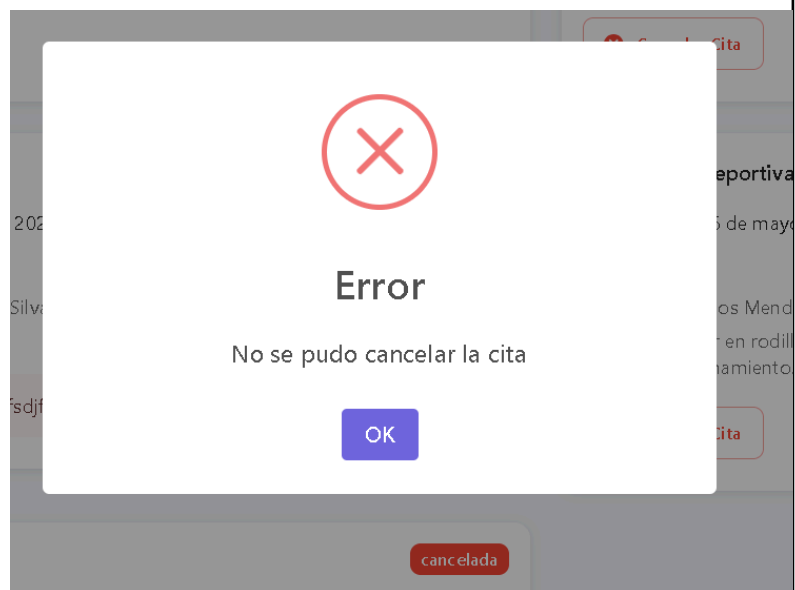
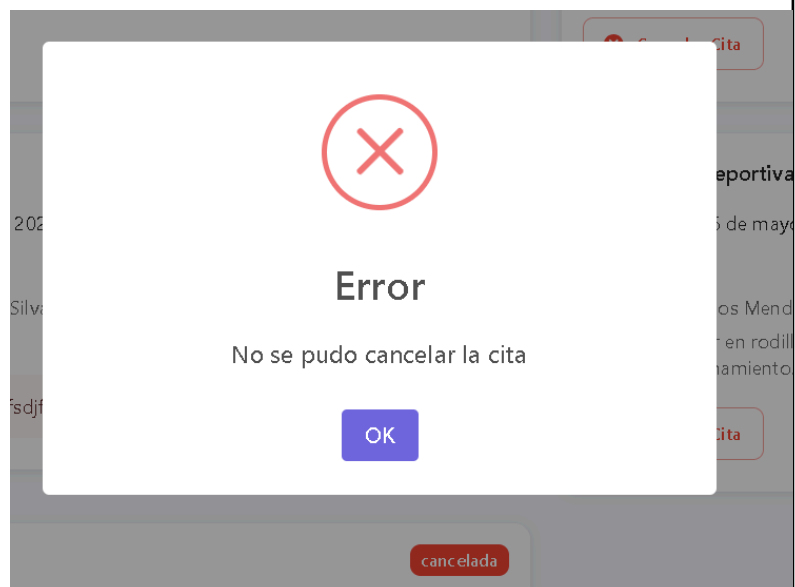
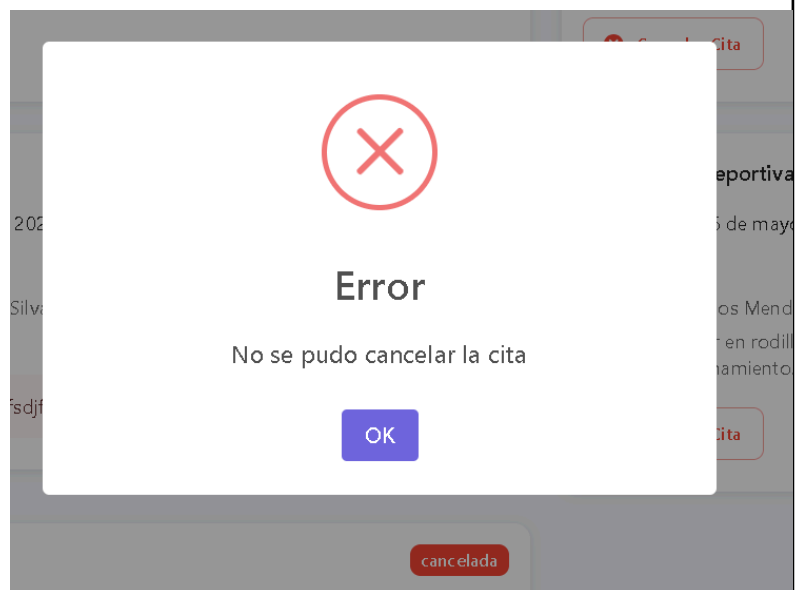
Walkthrough Review Status	
Author of Epic	Brayan Gualotuña (Lead Developer)
Reviewer	Isaac Erazo (Software Architect)
Status	Approved
Reviewer Comments	Test 1 (Bank Transfer Verification): Select bank transfer payment method. Actual Result: Passed. Status "Pending" and details displayed. Test 2 (Bank Transfer Verification): Upload receipt image or PDF. Actual Result: Passed. Visible for manual review. Test 3 (Bank Transfer Verification): Admin approves receipt. Actual Result: Passed. Status changes to "Confirmed". Test 4 (Edge Case): Upload unsupported file type. Actual Result: Passed. Validation error displayed. Test 5 (Cancellation Penalty): Cancel confirmed appointment >24 hours before. Actual Result: Passed. No penalty applied. Test 6 (Cancellation Penalty): Cancel confirmed appointment <24 hours before. Actual Result: Passed. 100% penalty warning shown. Test 7 (Financial Rule): Review revenue log after late cancellation. Actual Result: Passed. Recorded as earned penalty revenue. Test 8 (Mandatory Cancellation Reason): Cancel without entering reason. Actual Result: Passed. Action blocked. Test 9 (Mandatory Cancellation Reason): Reason shorter than 10 characters. Actual Result: Passed. Minimum length validation

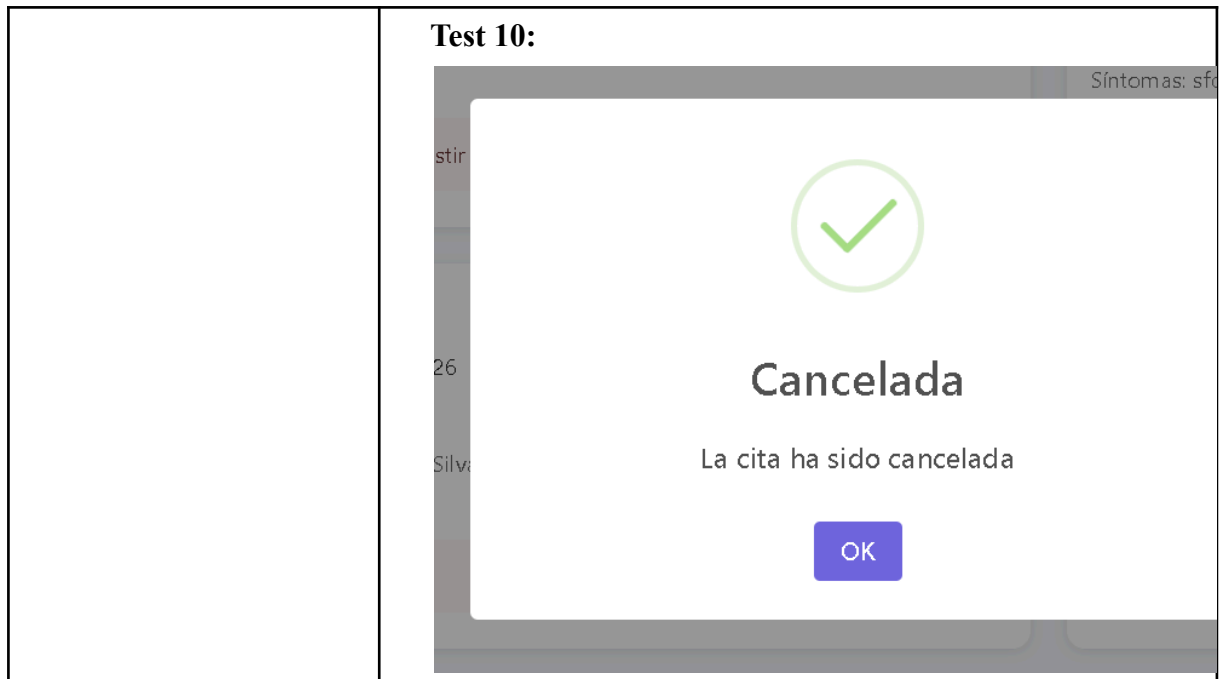
	appears. Test 10 (Audit): Cancel with valid reason >10 characters. Actual Result: Passed. Reason saved in audit record.
Evidence	Test 1: Próximas Citas Rehabilitación Post-Quirúrgica pendient 17 de mayo de 2026 09:30 Dr. Dr. Carlos Mendoza Silva Ver Detalles Test 2: Datos del Paciente Nombre Completo Juan Pérez García Cédula 1234567890 Teléfono 0987654321 Datos de la Cita Terapia Fisioterapia Deportiva Médico Dr. Carlos Mendoza Silva Fisioterapia Fecha jueves, 21 de mayo de 2026 Hora 12:00 Información Médica Síntomas sdfdsfdsafdf Exámenes Médicos No tiene exámenes Test 3: Con a 90 Confirmada La cita ha sido confirmada OK Test 4:

A button with a blue cloud icon and the text "Subir Exámenes".

Captura de pantalla 2025-07-09 105709.png
35.06 KB

Test 5:**Test 6:****Test 7:**

**Test 8:****Test 9:**



1.6 Epic 6: Session Packages & Digital Vouchers

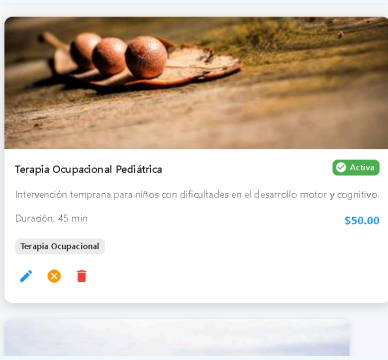
Description: Automates the mathematical tracking of prepaid therapies and generates professional outputs for patients.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
6.1	Automated Session Deduction	Desc: Tracks the remaining balance of pre-paid 10-session packages. AC: When a therapist marks a session "Completed", the patient's remaining sessions counter decrements by 1. Reaching 0 prompts a renewal alert.	5	High
6.2	Downloadable PDF Vouchers	Desc: Generate a digital receipt upon appointment confirmation. AC: PDF must include the Ser salud logo, Date, Time, Amount, Specialist, and a reminder to arrive 15 minutes early.	5	Med
6.3	Document Uploads	Desc: Allow patients to attach previous external lab results. AC: If digital files are unavailable, the UI must persistently remind the patient to bring physical copies to the first session.	3	Low

Walkthrough Review Status	
Author of Epic	Isaac Erazo (UI/UX)
Reviewer	César Galarza (Scrum Master)
Status	Still pending: implementing the PDF generator at the end of a quote and reviewing the financial system's allocation.
Reviewer Comments	Test 1 (Session Deduction): Therapist marks session as "Completed" for a patient with 10 sessions. Expected Result 1: Remaining sessions counter decrements by exactly 1 unit. Actual Result 1: Passed. Balance updates to 9.
	Test 2 (Session Deduction - Boundary): Attempt to deduct from a patient whose package balance is 0. Expected Result 2: System prevents deduction, throws 'Requires New Purchase' alert. Actual Result 2: Rejected; the payment system has not yet been implemented.
	Test 3 (Session Deduction - Input Validation): Force API to send -1 via Postman to deduct sessions. Expected Result 3: Backend rejects negative values and throws a 400 Bad Request error.

	● Actual Result 3: Passed.
	Test 4 (Session Deduction - Input Validation): Send the string "ONE" instead of an integer to the deduction endpoint. ● Expected Result 4: Type validation fails, preventing database corruption. ● Actual Result 4: Passed.
	Test 5 (PDF Vouchers): Click 'Download Digital Receipt' on a confirmed appointment. ● Expected Result 5: Generates PDF with clinic logo, Date, Time, Amount, and Specialist. ● Actual Result 5: Rejected; the PDF conversion system has not yet been implemented.
	Test 6 (PDF Vouchers - Edge Case): Browser strictly blocks pop-ups and downloads. ● Expected Result 6: A one slide-page fallback is generated on-screen. ● Actual Result 6: Passed.
	Test 7 (PDF Vouchers): Generate voucher for an appointment booked in the year 2099. ● Expected Result 7: PDF renders dates correctly without format overflowing. ● Actual Result 7: Passed, since selecting a date, the system only allows you to choose a date up to one week from now.
	Test 8 (Document Uploads): Upload a 6MB PDF laboratory result. ● Expected Result 8: File is rejected with an explicit clear UI error >5MB. ● Actual Result 8: Despite this, the system allowed a 16 MB PDF file to be uploaded without any warning.
	Test 9 (Document Uploads - Security): Rename a malicious .exe file to .pdf and attempt upload. ● Expected Result 9: Backend MIME-type checking

	intercepts and rejects the file. Actual Result 9: Passed.																														
	Test 10 (Document Uploads): Proceed through intake without attaching any files. Expected Result 10: Confirmation screen prints: "Please bring your physical medical documents to your first session." Actual Result 10: Passed.																														
Evidence	Test 1 (Session Deduction) SER SALUD Dashboard Terapias Mis Citas Perfil SER SALUD Gestiona tus citas y terapias desde aquí 2 Próximas Citas 6 Terapias Disponibles 0 Confirmadas 2 Pendientes Acciones Rápidas + Reservar Nueva Cita Ver Mis Citas Explorar Terapias Próximas Citas Fisioterapia Deportiva 15 de mayo de 2026 08:00 Dr. Dr. Carlos Mendoza Silva Ver Detalles Terapia Cognitivo- Conductual 15 de mayo de 2026 10:30 Dr. Dr. Roberto Silva Morales Ver Detalles Sesión iniciada como: Juan Pérez García Paciente																														
	Test 2 (Session Deduction - Boundary) SER SALUD Dashboard Gestión de Usuarios Gestión de Terapias Perfil SER SALUD Gestión de Usuarios Administra todos los usuarios del sistema 11 Total Usuarios 6 Pendientes 4 Médicos 1 Administradores Buscar por nombre, cédula Filtrar por Rol Todos los roles <table><tr><th>Nombre Completo</th><th>Cédula</th><th>Rol</th><th>Email</th><th>Teléfono</th><th>Acciones</th></tr><tr><td>Edison Lascano</td><td>1710893114</td><td>Paciente</td><td>edisonlascano@gmail.com</td><td>0961195050</td><td></td></tr><tr><td>César Luis Galarza Villalva</td><td>1754992541</td><td>Paciente</td><td>dwer.cesar123@gmail.com</td><td>0995572799</td><td></td></tr><tr><td>David Rodriguez</td><td>1726440538</td><td>Paciente</td><td>david@gmail.com</td><td>0958475689</td><td></td></tr><tr><td>Dra. Ana Martínez Vega</td><td>4444444444</td><td>Médico</td><td>ana.martinez@fiova.com</td><td>0994444444</td><td></td></tr></table> Sesión iniciada como: Administrador del Sistema Administrador	Nombre Completo	Cédula	Rol	Email	Teléfono	Acciones	Edison Lascano	1710893114	Paciente	edisonlascano@gmail.com	0961195050		César Luis Galarza Villalva	1754992541	Paciente	dwer.cesar123@gmail.com	0995572799		David Rodriguez	1726440538	Paciente	david@gmail.com	0958475689		Dra. Ana Martínez Vega	4444444444	Médico	ana.martinez@fiova.com	0994444444	
Nombre Completo	Cédula	Rol	Email	Teléfono	Acciones																										
Edison Lascano	1710893114	Paciente	edisonlascano@gmail.com	0961195050																											
César Luis Galarza Villalva	1754992541	Paciente	dwer.cesar123@gmail.com	0995572799																											
David Rodriguez	1726440538	Paciente	david@gmail.com	0958475689																											
Dra. Ana Martínez Vega	4444444444	Médico	ana.martinez@fiova.com	0994444444																											
	Test 3 (Session Deduction - Input Validation)																														



Storage

Local storage

Session storage

Extension storage

Cookies

Private state tokens

Interest groups

Shared storage

Cache storage

Storage buckets

Background services

Backforward cache

Background fetch

Background sync

Bounce tracking mtl...

Notifications

Payment handler

Periodic background ...

Speculative loads

Push messaging

Background ABI

Key	Value
auth_token	eyJhbGciOiJIUzI1NiIsInR5c...
selectedTerapia	{\"id\":15,\"nombre\":\"Terapia Oc...

(Id: 15, nombre: \"Terapia Ocupacional Pediátrica\",...

activo: true

createdAt: \"2026-05-04T15:44:25.567Z\"

descripcion: \"Intervención temprana para niños c...

duracion: -1

especialidad: \"Terapia Ocupacional\"

id: 15

image: \"https://picsum.photos/seed/fo14688/300\"

nombre: \"Terapia Ocupacional. Pediatría\"

precio: 58

updatedAt: \"2026-05-04T15:44:25.567Z\"

Test 4 (Session Deduction - Input Validation)

Nueva Terapia

Nombre de la Terapia

Descripción

Especialidad

Duración (minutos)

0

Precio (USD)

\$ 0

URL de Imagen (opcion...

0.18

0.5

0.16

0.15

Cancelar

Crear

Test 5 (PDF Vouchers)

Detalles de la Cita

Terapia Cognitivo-Conductual

pendiente

Fecha

15 de mayo de 2026

Hora

10:30

Médico

Dr. Dr. Roberto Silva Morales

Sintomas

sFDSAFDSFSD

Exámenes Médicos

No tiene exámenes

Duración

60 minutos

Precio

\$55.00

Cerrar

Ver Todas Mis Citas

Test 6 (PDF Vouchers - Edge Case)

SER SALUD

Dashboard

Terapias

Mis Citas

Perfil

SER SALUD

Selección un Médico

Elige el médico especialista para tu terapia de Fisioterapia Deportiva

Fisioterapia Deportiva

Fisioterapia • 60 min • \$45.00

Médicos Disponibles (2)

D

Dr. Carlos Mendoza Silva

Fisioterapia

Lic. MED-FIS-001

★★★★★ 4.8 • 245 pacientes

Horario de Atención

Lun, Mar, Mié, Jue, Vie, Sáb

08:00 - 16:00

D

Dra. Ana Martínez Vega

Fisioterapia

Lic. MED-FIS-004

★★★★★ 4.6 • 156 pacientes

Horario de Atención

Lun, Mar, Mié, Jue, Vie

14:00 - 20:00

Sesión iniciada como:

Juan Pérez García

Paciente

Continuar al Calendario

Test 7 (PDF Vouchers)

SER SALUD

Selección Fecha y Hora

Terapia seleccionada

Fisioterapia Deportiva

60 min • \$45.00

Médico seleccionado

Dra. Ana Martínez Vega

Fisioterapia

1. Selección una Fecha

vie

15

may

sáb

16

may

dom

17

may

lun

18

may

mar

19

may

mié

20

may

jue

21

may

2. Selección una Hora

Primero selecciona una fecha

Test 8 (Document Uploads)

¿Tienes exámenes médicos?

☒ Sí

☐ No

Subir Exámenes

HTML_UserStories_SerSalud.pdf

242.79 KB

Chavez_Chuqui_Galarza_Guasumba_ODS original antes de mi.pdf

16304.37 KB

Si no los tienes en digital, no olvides llevarlos el día de tu cita

Continuar a Confirmación →

Test 9 (Document Uploads - Security)

	¿Tienes exámenes médicos? ☒ Sí ☐ No Subir Exámenes Si no los tienes en digital, no olvides llevarlos el día de tu cita Continuar a Confirmación →
	Test 10 (Document Uploads) ¿Tienes exámenes médicos? ☒ Sí ☐ No Subir Exámenes Si no los tienes en digital, no olvides llevarlos el día de tu cita Continuar a Confirmación →

1.7 Epic 7: Administrative Configuration & System Scalability

Description: Provides the Customer (Clinic Owner) and administrative staff with total autonomy to scale the business without requiring further developer intervention, allowing the dynamic addition of new medical branches and personnel.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
7.1	Dynamic Service Creation	Desc: Allows the Admin to create new therapy categories (e.g., Psychology, Nutrition) dynamically to expand the clinic's offerings. AC: The new service immediately populates the public frontend dropdown menu. The system strictly prevents service creation if a responsible specialist is not assigned.	5	High

7.2	Staff Roster Management	Desc: Interface for adding, suspending, or updating specialist profiles and departments. AC: Suspended specialists are immediately hidden from the public booking engine, but their historical session data remains intact in the database for auditing.	5	Med
7.3	Global Variables Configuration	Desc: Admin module to update the clinic's global variables, such as the WhatsApp emergency number or official Bank Account. AC: Changes reflect instantly across all patient-facing interfaces and payment check-out screens without requiring a server reboot.	3	Low

Walkthrough Review Status	
Author of Epic	César Galarza (Scrum Master)
Reviewer	Brayan Gualotuña (Lead Developer)
Status	A
Reviewer Comments	A
Evidence	

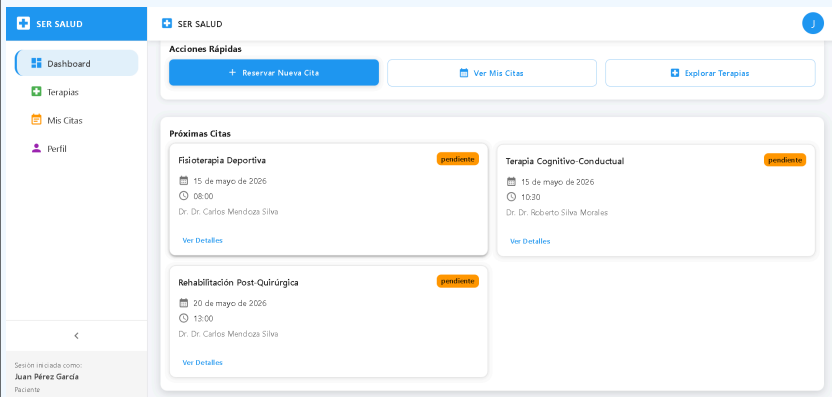
1.8 Epic 8: Financial Analytics & Operational Metrics

Description: Transforms raw scheduling data into actionable business intelligence, allowing the Customer to monitor the clinic's financial health, liability in prepaid sessions, and overall staff efficiency.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
8.1	Financial Liability Dashboard	Desc: A visual analytics tab calculating the clinic's exact financial debt regarding prepaid but unfulfilled therapy sessions. AC: Aggregates all 50% advance payments of 'Pending' and 'Confirmed' upcoming sessions into a distinct 'Liabilities' visual chart.	8	High
8.2	Penalty Revenue Tracking	Desc: Automated financial tracking of deposits retained due to the 24-hour cancellation rule. AC: When a late cancellation occurs, the system automatically subtracts the deposit from the 'Liability' column and adds it to an 'Earned Penalty Revenue' ledger.	5	High
8.3	Clinic Occupation Rate	Desc: Evaluates the efficiency of the special-ists by comparing booked hours versus available working hours. AC: Displays a dynamic Pie Chart to the Admin, filtering out manually blocked hours (like lunch breaks) so they do not negatively skew the efficiency percentage.	5	Med

Walkthrough Review Status	
Author of Epic	Isaac Erazo (Software Architect)
Reviewer	César Galarza (Scrum Master)
Status	Completely rejected, the system still does not include any features related to on-site financial calculations.
Reviewer Comments	Test 1 (Financial Liability): Process 5 pending appointments with a \$50 advance each. Expected Result 1: Visual chart aggregates advances and displays a \$250 liability debt. Actual Result 1: Rejected.
	Test 2 (Financial Liability - Extreme Value): Input a manual payment advance of \$10000000. Expected Result 2: Chart scales dynamically without

	breaking the UI CSS container. ● Actual Result 2: Rejected.
	Test 3 (Financial Liability - Null Test): Simulate a database entry with a missing/null advance value. ● Expected Result 3: Dashboard treats null as \$0 and calculates the rest without throwing a TypeError. ● Actual Result 3: Rejected.
	Test 4 (Financial Liability): Cancel an appointment with <24h notice. ● Expected Result 4: Value is automatically moved from 'Liabilities' to 'Earned Revenue'. ● Actual Result 4: Rejected.
	Test 5 (Penalty Tracking): Execute a late cancellation (triggering the 100% penalty rule). ● Expected Result 5: System adds the exact deposit amount to the 'Earned Penalty Revenue' ledger. ● Actual Result 5: Rejected.
	Test 6 (Penalty Tracking): Execute a normal cancellation (>24h notice). ● Expected Result 6: Penalty ledger remains completely unchanged. ● Actual Result 6: Passed.
	Test 7 (Penalty Tracking): Admin performs a manual override/refund of a penalized session. ● Expected Result 7: Penalty revenue is accurately deducted. ● Actual Result 7: Rejected.
	Test 8 (Occupation Rate): Filter the efficiency chart for the current week. ● Expected Result 8: Pie chart filters out manually blocked hours (e.g., lunch breaks) from the calculation. ● Actual Result 8: Rejected.

	Test 9 (Occupation Rate - Boundary): Check the dashboard on a week with zero bookings. ● Expected Result 9: Displays 0% occupation without throwing a "Division by Zero" backend crash. ● Actual Result 9: Rejected.
	Test 10 (Occupation Rate - Boundary): Fully book every single available slot for a therapist. ● Expected Result 10: Displays exactly 100% occupation without decimal rounding errors. ● Actual Result 10: Rejected.
Evidence	 The screenshot displays the 'SER SALUD' dashboard. On the left is a sidebar menu with 'Dashboard' selected, and other options like 'Terapias', 'Mis Citas', and 'Perfil'. The main content area is titled 'Acciones Rápidas' and includes buttons for '+ Reservar Nueva Cita', 'Ver Mis Citas', and 'Explorar Terapias'. Below this, the 'Próximas Citas' section shows three upcoming appointments: 'Fisioterapia Deportiva' (May 15, 2026, 08:00, Dr. Carlos Mendoza Silva), 'Terapia Cognitivo-Conductual' (May 15, 2026, 10:30, Dr. Roberto Silva Morales), and 'Rehabilitación Post-Quirúrgica' (May 20, 2026, 12:00, Dr. Carlos Mendoza Silva). Each appointment card has a 'pendiente' status tag and a 'Ver Detalles' link. At the bottom left, a user session notice indicates the user is logged in as 'Juan Pérez García'.

1.9 Epic 9: Clinical Document Management

Description: Focuses on the secure handling of external medical files and the internal logging of post-therapy observations to build a comprehensive, cross-departmental patient record.

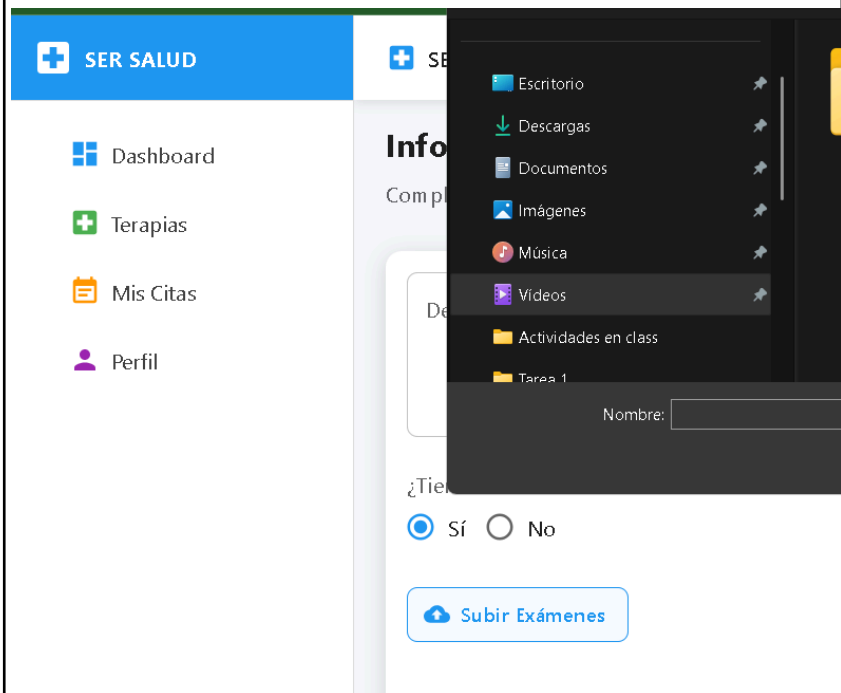
ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
9.1	External Exam Up-loads	Desc: Allows patients to securely attach lab-oratory results, X-Rays, or doctor referrals (PDF/JPG) during the intake phase. AC: Files over 5MB are rejected with a clear UI error. Uploaded files are directly linked to the specific appointment ID.	8	Med
9.2	Physical Document Reminder	Desc: A persistent UI nudge for patients who bypass the digital document upload phase. AC: If no file is attached, the confirmation screen and the PDF Voucher must explicitly print: "Please bring your physical medical documents to your first session."	3	Low
9.3	Post-Session Observation Ledger	Desc: A dedicated text area for therapists to log treatment details, patient progress, and specific physical reactions after a session. AC: This ledger is permanently attached to the patient's ID and is fully visible to other specialists if the patient is internally referred to another department.	5	High

Walkthrough Review Status	
Author of Epic	Brayan Gualotuña (Lead Developer)
Reviewer	Isaac Erazo (Software Architect)
Status	Approved
Reviewer Comments	Test 1 (External Exam Uploads): Upload PDF medical exam. Actual Result: Passed. Linked to appointment ID. Test 2 (External Exam Uploads): Upload JPG X-Ray. Actual Result: Passed. Displayed in document area. Test 3 (External Exam Uploads): Upload file >5MB. Actual Result: Passed Test 4 (Security): Upload executable renamed as document. Actual Result: Didn't pass, didn't prevent it Test 5 (Physical Document Reminder): Confirmation without

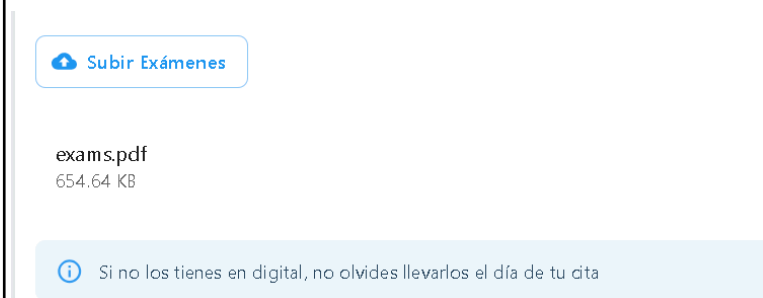
	upload. Actual Result: Passed. Reminder displayed. Test 6 (Physical Document Reminder): Generate PDF voucher skipping upload. Actual Result: Passed. Voucher includes reminder text. Test 7 (Edge Case): Upload then remove document before confirmation. Actual Result: Passed. Reminder restored. Test 8 (Post-Session Observation Ledger): Therapist enters treatment notes. Actual Result: Passed. Saved permanently under patient ID. Test 9 (Post-Session Observation Ledger): Refer patient to another department. Actual Result: NOT IMPLEMENTED YET. Test 10 (Data Integrity): Edit new observation with existing history. Actual Result: NOT IMPLEMENTED YET
--	---

Evidence

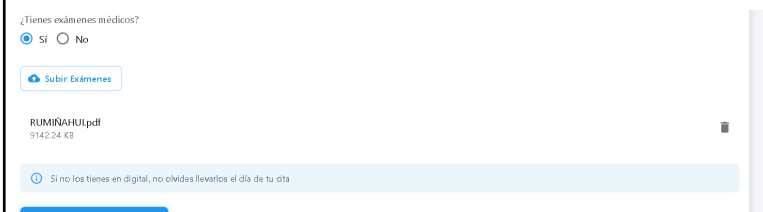
Test 1:



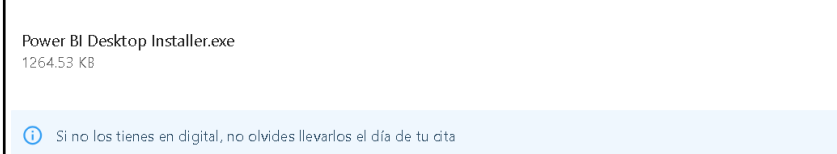
Test 2:



Test 3:



Test 4:



Test 5:

Datos del Paciente

Nombre Completo
Juan Pérez García

Cédula
1234567890

Teléfono
0987654321

Datos de la Cita

Terapia
Rehabilitación Post-Quirúrgica

Médico
Dr. Carlos Mendoza Silva

Fisioterapia

Fecha
domingo 17 de mayo de 2026

Información Médica

Síntomas
dolor de pie

Exámenes Médicos
No tiene exámenes

Test 6:

Rehabilitación Post-Quirúrgica

pendiente

📅

domingo, 17 de mayo de 2026

🕒

09:30

Médico: Dr. Carlos Mendoza Silva

Síntomas: dfdasfdssadfsad

✖

 Cancelar Cita

Test 7:

📌 Si no los tienes en digital, no olvides llevarlos el día de tu cita

Test 8:

Completar cita

Notas de la consulta (opcional)

The treatment was successful

Completar Cancelar

1.10 Epic 10: Patient Engagement & Automated Flow

Description: Implements gamification strategies and automated reminders to optimize the clinic's time, reducing the administrative burden on the staff.

ID	PBI Title	Description & Acceptance Criteria	Pts	Pri
10.1	Profile Completion Incentive	Desc: A gamified prompt to encourage patients to fill out the dense medical history from home, saving 30 minutes of clinical time. AC: The dashboard displays a banner offering a predefined discount or benefit on their next package if they complete their profile to 100% digitally.	5	Low
10.2	Pre-Appointment Reminders	Desc: System automation to ensure patients do not forget their scheduled and confirmed therapies. AC: A cron job triggers an automated alert 12 hours prior to the session, explicitly re-minding the patient to arrive 15 minutes early to optimize clinical flow.	5	Med
10.3	Post-Therapy Satisfaction Nudge	Desc: A brief feedback mechanism to ensure quality of service and patient retention. AC: 24 hours after a completed session, the system prompts the user in their dashboard to rate their experience out of 5 stars, flagging any 1-star reviews for the Admin.	3	Low

Walkthrough Review Status	
Author of Epic	Isaac Erazo (UI/UX)
Reviewer	César Galarza (Scrum Master)
Status	While partially accepted, the system still needs to be integrated with third-party APIs for automation and approved security measures.
Reviewer Comments	Test 1 (Pre-Appointment Reminders): System checks cron job for upcoming appointments. • Expected Result 1: Triggers alert exactly 12 hours prior to the session. • Actual Result 1: Passed.
	Test 2 (Pre-Appointment Reminders): Verify the content of the automatization. • Expected Result 2: Explicitly reminds the patient to arrive 15 minutes early. • Actual Result 2: Passed.

	Test 3 (Pre-Appointment Reminders - Edge Case): Book an appointment with exactly 12.5 hours of notice. ● Expected Result 3: Cron job picks up the appointment in its next cycle and sends the reminder immediately. ● Actual Result 3: Passed.
	Test 4 (Pre-Appointment Reminders - Error Handling): User profile contains a malformed email address. ● Expected Result 4: Email bounce is caught, cron job continues processing the rest of the queue without failing entirely. ● Actual Result 4: Rejected, the system does not yet implement APIs or libraries from third-party companies for validation via Google or email.
	Test 5 (Profile Incentive): User accesses dashboard with an incomplete medical history. ● Expected Result 5: Dashboard displays banner offering predefined discount for 100% completion. ● Actual Result 5: Rejected, there is no option to access the medical history form for manual completion; it can only be filled out as a PDF.
	Test 6 (Profile Incentive): User completes profile to 100%. ● Expected Result 6: Banner disappears dynamically and backend applies discount flag to the account. ● Actual Result 6: Rejected: There is no option to access the medical history form for manual completion; it can only be filled out as a PDF.
	Test 7 (Profile Incentive - Security): User attempts to manually inject profile_complete=true in local storage/cookies. ● Expected Result 7: Backend validates missing fields and ignores the manipulated frontend state. ● Actual Result 7: Passed.
	Test 8 (Satisfaction Nudge): User logs in 24 hours after a completed session. ● Expected Result 8: System prompts user to rate

	experience out of 5 stars. ● Actual Result 8: Rejected.
--	---

Selecciona Fecha y Hora

Terapia seleccionada

Fisioterapia Deportiva

60 min • \$45.00

Médico seleccionado

Dr. Carlos Mendoza Silva

Fisioterapia

1. Selecciona una Fecha

we

sáb

dom

lun

mar

mié

jue

15

16

17

18

19

20

21

may

may

may

may

may

may

may

2. Selecciona una Hora

08:30

09:00

09:30

10:00

10:30

11:00

11:30

Resumen de tu Selección

Fecha: 15/5/2026

Hora: 08:30

Continuar al Formulario →

Test 3 (Pre-Appointment Reminders - Edge Case)

Selecciona Fecha y Hora

Terapia seleccionada

Fisioterapia Deportiva

60 min • \$45.00

Médico seleccionado

Dr. Carlos Mendoza Silva

Fisioterapia

1. Selecciona una Fecha

we

sáb

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mié

jue

15

16

17

18

19

20

21

may

may

may

may

may

may

may

2. Selecciona una Hora

08:30

09:00

09:30

10:00

10:30

11:00

11:30

Resumen de tu Selección

Fecha: 15/5/2026

Hora: 08:30

Continuar al Formulario →

Test 4 (Pre-Appointment Reminders - Error Handling)

J

Juan Pérez García

Paciente

Cédula: 1234567890

Información Personal

Nombre Completo

Juan Pérez García

Email

juan.perez@email.com

Teléfono

0987654321

Edad

No especificado años

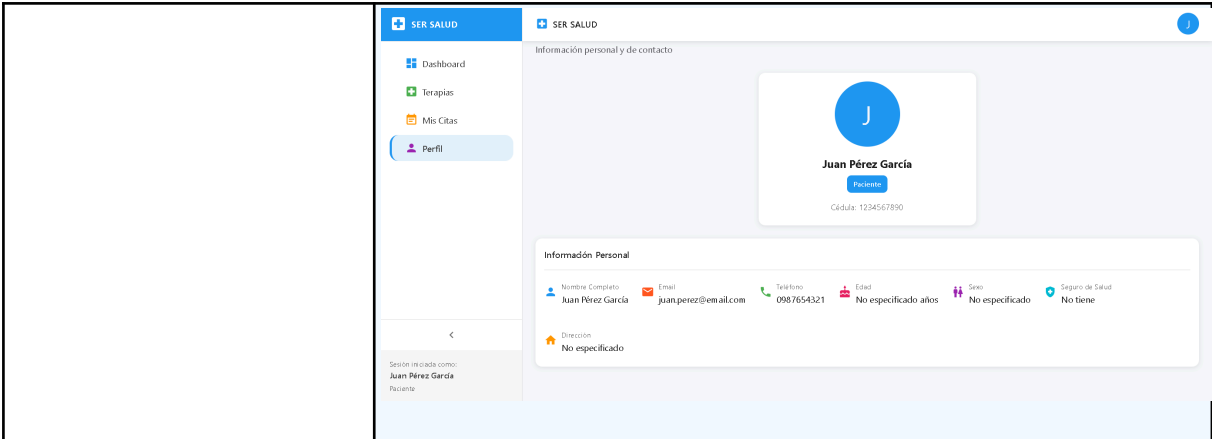
Sexo

No especificado

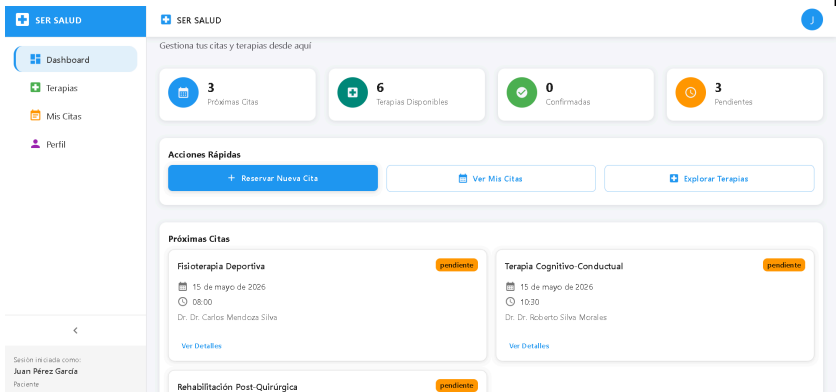
Dirección

No especificado

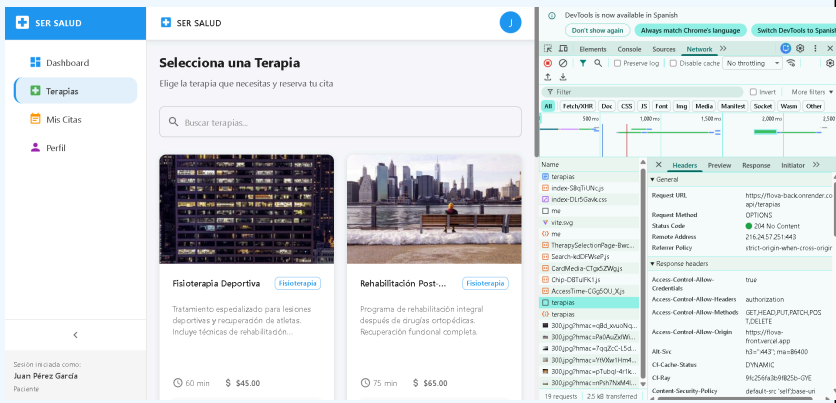
Test 5 (Profile Incentive)



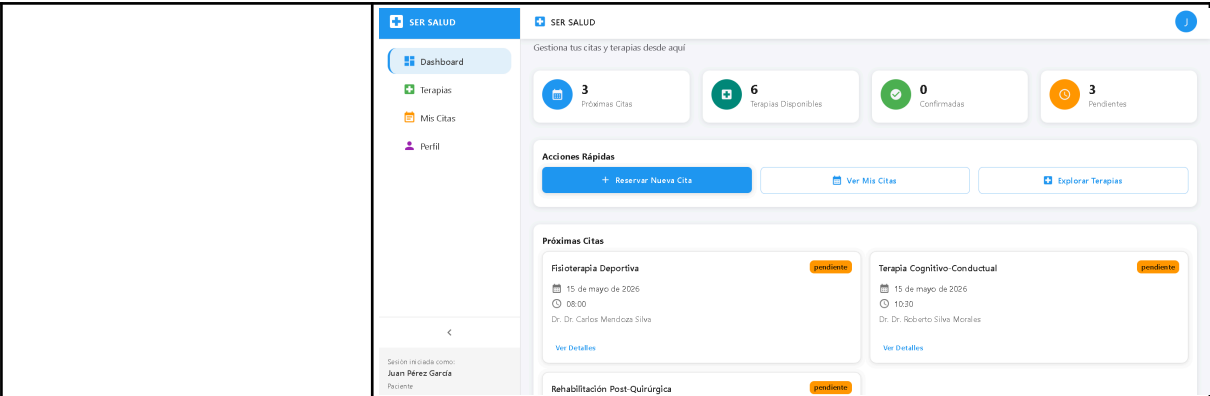
Test 6 (Profile Incentive)



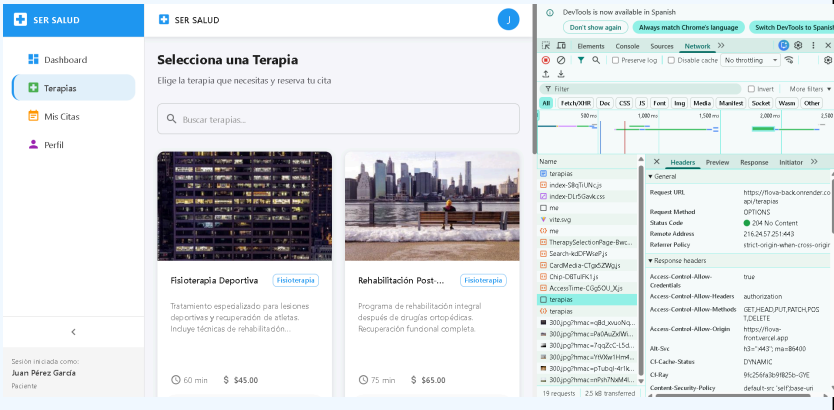
Test 7 (Profile Incentive - Security)



Test 8 (Satisfaction Nudge)



Test 9 (Satisfaction Nudge)



Test 10 (Satisfaction Nudge - Security)

